

Observation Theory of Mobility Inference: Fisher Information Sufficiency in Aggregate Travel-Distance Representations

Thinh Nguyen^{1*}

^{1*} Aarista Technologies, Ho Chi Minh City, Vietnam.

Corresponding author(s). E-mail(s): nqt900@gmail.com;

Abstract

Accurate spatial interaction modeling traditionally relies on complete origin–destination (OD) flow observations, operating under the implicit *Data Richness Paradigm* that disaggregate trajectory data are indispensable for mobility inference. Challenging this paradigm, this study formulates a formal *Observation Theory of Mobility Inference*. We establish the *Observation Sufficiency Principle*: an observation space $\mathcal{H}$ is valid for identifying the parameters θ of a spatial interaction process $\mathcal{M}(\theta)$ if and only if its image under an observation operator $\mathcal{P}$ preserves sufficient Fisher information to uniquely identify θ . Structuring the theory from general distance-deterrence models down to Tanner gravity parameterizations, we derive a 4-stage *Statistical Representation Layer* connecting spatial interaction processes to aggregate distance histograms. Deductively derived theoretical predictions assert that aggregate distance representations induce equivalent statistical evidence for parameter identification compared to complete OD matrices. Empirical validation across 50 US metropolitan areas confirms these predictions: distance-decay parameters identified from aggregate summaries match those calibrated on complete OD matrices, and downstream spatial interaction predictions achieve statistical equivalence within a 0.01 Common Part of Commuters (CPC) margin. A game-theoretic Shapley analysis confirms that recovered distance friction accounts for 85.8% of total predictive performance. Rather than asking how to reconstruct point-to-point flows from big data, this work establishes what information is fundamentally required to identify spatial interaction models.

Keywords: Urban mobility, observation theory, Fisher information, distance decay calibration, gravity models, statistical representation layer, parameter identification

1 Introduction

1.1 Big Scientific Problem

Accurate origin–destination (OD) flow estimation is fundamental to transportation planning, infrastructure prioritization, and urban mobility policy analysis (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). To model spatial interaction, spatial interaction frameworks such as gravity models (Wilson 1971; Tanner 1961) rely on distance-decay parameters to capture

the behavioral friction of travel distance. Recent learning-based frameworks, including DeepGravity and graph neural networks (Simini et al. 2021; Hu et al. 2020), similarly depend on estimating spatial friction representations from observed data.

However, existing mobility inference approaches share a fundamental dependency: they operate under an implicit **Data Richness Paradigm**, assuming that complete origin–destination observations or disaggregate travel

trajectories are indispensable for parameter calibration or model training. Acquiring complete OD observations remains a major practical challenge. Household travel surveys require substantial financial resources, limiting their coverage to a small fraction of cities worldwide (Egu and Bonnel 2020; Hussain et al. 2021). Mobile-phone and GPS trajectory records, while richer in spatial detail, are frequently unavailable due to commercial licensing restrictions and strict privacy regulations (de Montjoye et al. 2013). As a result, complete OD observations are difficult to obtain for the majority of urban regions globally. Meanwhile, aggregate mobility products—including Meta’s Movement Distribution Maps (MDM) (Meta 2026)—are becoming increasingly available across platforms and regions, providing privacy-preserving summaries of population travel behavior without revealing origin–destination pairs.

1.2 Research Gap

Existing work has focused on reconstructing or predicting mobility flows from alternative data sources or transferring learned representations from data-rich regions. However, it remains unknown whether aggregate travel-distance distributions preserve sufficient statistical information for identifying the parameters of distance-deterrence mobility models.

1.3 Research Question

To address this gap, this paper addresses one central research question:

Can aggregate travel-distance distributions replace complete OD observations for identifying the parameters of distance-deterrence mobility models?

1.4 Central Hypothesis

We evaluate the following central hypothesis:

Aggregate travel-distance distributions constitute a statistical representation of the underlying spatial interaction process that preserves sufficient Fisher information to identify latent distance-decay parameters.

1.5 Contributions

The contributions of this study establish a formal Observation Theory:

1. **Formulation of the Observation Theory Framework:** We formalize the Observation Sufficiency Principle, defining an observation operator $\mathcal{P}$ mapping spatial interaction space onto a Statistical Representation Layer.
2. **Deductive Theoretical Derivations:** We derive the Fisher Information backbone and aggregate multinomial log-likelihood, proving that maximum likelihood estimation over distance fractions is optimized via cross-entropy.
3. **Demonstration of Statistical Evidence Equivalence:** We demonstrate empirically across 50 metropolitan areas that aggregate distance representations induce essentially the same statistical evidence for parameter identification as complete OD matrices.
4. **Demonstration of Preservation of Predictive Consequences:** We demonstrate that parameters identified under aggregate representations preserve downstream predictive capability in spatial interaction modeling, accounting for 85.8% of total flow prediction accuracy.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 formalizes Observation Theory, the Observation Sufficiency Principle, the Statistical Representation Layer, and Fisher Information derivations. Section 4 evaluates the theoretical predictions. Section 5 explores the Information Sufficiency Paradigm, conceptual negative results, explicit limitations, and the multi-paper Research Program, and Section 6 concludes the paper.

2 Related Work

2.1 Mobility Modeling under Limited Behavioral Information

Accurate origin–destination (OD) flow modeling is a core task in spatial interaction analysis, travel demand forecasting, and urban policy evaluation. Traditionally, OD matrices are constructed from household travel surveys, census commuting records, mobile-phone trajectories, or GPS traces. While these data sources provide detailed mobility observations, they are expensive to collect, infrequently updated, and increasingly constrained by privacy regulations (de Montjoye et al. 2013). Consequently, acquiring complete OD observations

Table 1 Comparative analysis of spatial interaction approaches across modeling paradigms, statistical representation layers, and observation requirements.

Model Class	Decomposed Structure	Transferable Across Cities	Requires Target OD	Interpretable	Decay Explicitly Modeled
Gravity (Wilson 1971)	Yes	No	Yes	High	Yes
Radiation (Simini et al. 2012)	No	Yes	No	High	Yes
DeepGravity (Simini et al. 2021)	No	Limited	Yes	Low	No
Graph-based OD (MPGCN) (Hu et al. 2020)	No	Limited	Yes	Low	No
Graph Transfer Learning (GODDAG) (Rong et al. 2023a)	Partial	Yes	Yes	Low	No
Gravity-Informed Neural Models (Rong et al. 2023b)	Partial	Limited	Yes	Medium	Partial
Proposed Observation Framework	Yes	Yes	No	High	Yes

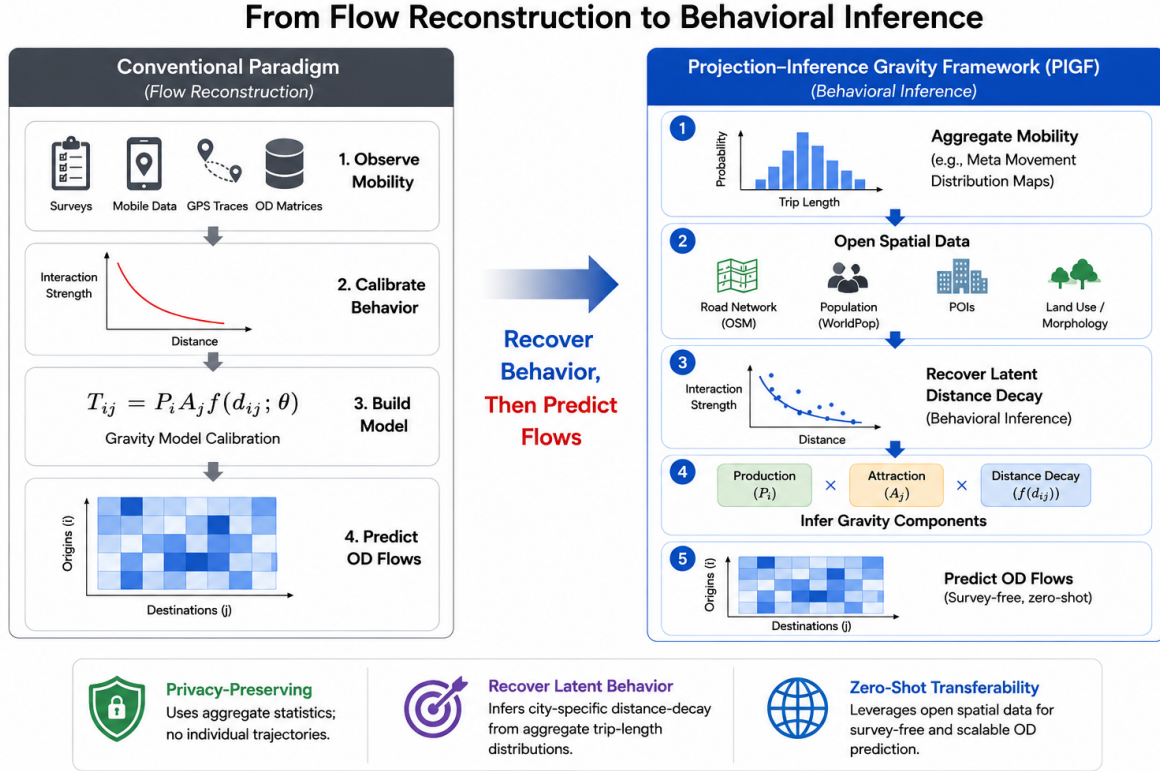


Fig. 1 Conceptual distinction between direct flow reconstruction paradigms and the proposed parameter inference paradigm under Observation Theory. Rather than attempting to reconstruct point-to-point mobility matrices from complete OD surveys, the proposed framework infers latent distance-decay parameters directly from aggregate travel-distance representations prior to downstream spatial interaction generation.

remains a major bottleneck for spatial interaction modeling, particularly in data-scarce regions.

To address data scarcity, classical spatial interaction models estimate mobility through analytical formulations (Wilson 1971; Simini et al. 2012), whereas recent deep learning methods exploit spatial representations learned from historical mobility data (Simini et al. 2021; Enaya et al. 2026). Despite methodological differences, existing approaches share a common dependency: they

rely on observed OD flows or individual trajectories to estimate distance decay or to train spatial interaction models (Lenormand et al. 2016; Simini et al. 2021). Even recent transfer-learning and zero-shot frameworks assume that representative OD observations are available in source domains (Yang et al. 2026).

2.2 Distance Decay in Spatial Interaction

Distance decay forms the core behavioral constraint in spatial interaction models, describing how interaction probability diminishes with travel impedance (Tanner 1961; Wilson 1971; Fotheringham and O’Kelly 1989; Sen and Smith 1995). Heterogeneous travel behavior is commonly represented using functional forms such as exponential, power-law, or Tanner deterrence functions. The two-parameter Tanner function is widely adopted because it simultaneously captures short-distance concentration and long-distance attenuation.

Existing distance-decay calibration methods, however, assume the availability of observed origin–destination matrices to fit parameters via maximum-likelihood or least-squares estimation (Flowerdew and Aitkin 1982; Lenormand et al. 2016; Rubio-Herrero and Muñuzuri 2023). Consequently, methodological efforts have focused primarily on how to fit deterrence functions given complete OD data, rather than investigating whether deterrence parameters can be identified without observed OD matrices.

2.3 Information Content of Aggregate Mobility Summaries

Privacy-preserving aggregate mobility products—such as Meta Movement Distribution Maps (MDM) (Meta 2026)—summarize population travel behavior into trip-length distributions (TLDs) while discarding origin–destination identities (Oliver et al. 2020; Buckee et al. 2020; Pappalardo et al. 2023). Previous studies have used aggregate mobility statistics primarily for descriptive mobility analysis, epidemic monitoring, and accessibility assessment.

Although aggregate mobility summaries omit location identities, they retain the empirical distribution of travel distances across an urban region. Whether aggregate distance distributions preserve sufficient statistical information to identify distance-decay parameters of underlying spatial interaction models remains uncharacterized, defining the central research gap addressed in this study.

3 Observation Theory and Probabilistic Model

This section formalizes Observation Theory, introducing its mathematical objects, the Observation Sufficiency Principle, the Statistical Representation Layer, and the Fisher Information backbone.

3.1 Hierarchical Formulation and Objects of Observation Theory

To establish Observation Theory as a general theoretical framework, we structure it hierarchically from general principles to specific parameterizations:

$$\text{Observation Theory} \rightarrow \text{Distance Decay} \rightarrow \text{Gravity} \rightarrow \text{Tanner}. \quad (1)$$

Observation Theory is defined by five foundational mathematical objects:

1. **Latent Process** ($\mathcal{M}(\theta)$): The unobserved spatial interaction process governed by parameter vector $\theta \in \Theta$.
2. **Observation Space** ($\mathcal{H}$): The observable measurement domain (e.g., discrete distance histograms across K intervals).
3. **Observation Operator** ($\mathcal{P}$): The linear projection operator $\mathcal{P} : \mathcal{M} \rightarrow \mathcal{H}$ mapping full spatial interaction space onto the observation space:

$$\mathcal{P}(\mathbf{T})_b = \sum_{i=1}^{N_{\text{zones}}} \sum_{j=1}^{N_{\text{zones}}} T_{ij} \cdot \mathbf{1}(d_{ij} \in b). \quad (2)$$

4. **Statistical Representation Layer** ($\mathbf{b}$): The normalized probability distribution vector $\mathbf{b} = (b_1, \dots, b_K)^\top \in \mathcal{H}$ induced by operator $\mathcal{P}$.
5. **Parameter Estimator** ($\hat{\theta}$): The inference mapping $\hat{\theta} : \mathcal{H} \rightarrow \Theta$ recovering governing parameters from the representation layer.

3.2 The Observation Sufficiency Principle

We state the fundamental principle of Observation Theory:

Observation Sufficiency Principle: *An observation space $\mathcal{H}$ is a valid observation space for identifying the parameter vector θ of a spatial interaction process $\mathcal{M}(\theta)$ under observation operator $\mathcal{P}$*

if and only if the image $\mathcal{P}(\mathcal{M})$ preserves sufficient Fisher information to uniquely identify θ .

Under this principle, an observation (whether complete OD matrix, GPS trajectory, or distance histogram) is not defined by its physical storage format, but as a mapping $\mathcal{P} : \mathcal{M} \rightarrow \mathcal{H}$. If the mapping preserves Fisher information $\mathcal{I}(\theta)$, parameter inference is mathematically guaranteed.

3.3 Fisher Information Backbone

Let $\log \mathcal{L}(\theta; \mathbf{y})$ denote the log-likelihood of observing representation vector $\mathbf{y} \in \mathcal{H}$. The ****Fisher Information Matrix**** $\mathcal{I}(\theta) \in \mathbb{R}^{p \times p}$ is defined as:

$$\mathcal{I}(\theta) = -\mathbb{E}_{\mathbf{y} \sim \pi_\theta} [\nabla_\theta^2 \log \mathcal{L}(\theta; \mathbf{y})]. \quad (3)$$

Parameter vector θ is strictly identifiable from observation space $\mathcal{H}$ if and only if the Fisher Information Matrix $\mathcal{I}(\theta)$ is non-singular ($\det \mathcal{I}(\theta) > 0$).

3.4 Theoretical Predictions Derived from the Principle

From the Observation Sufficiency Principle, three theoretical predictions follow deductively:

- **Prediction 1 (Existence of Probabilistic Observation Model):** The projection $\mathcal{P}(\mathcal{M})$ yields a discrete probability distribution $\pi(\theta)$ that induces a well-defined Multinomial log-likelihood whose Fisher information matrix $\mathcal{I}(\theta)$ is non-singular.
- **Prediction 2 (Statistical Evidence Equivalence):** If observation space $\mathcal{H}$ preserves sufficient Fisher information, parameter estimates $\hat{\theta}_{\mathcal{H}}$ identified from $\mathcal{H}$ induce equivalent statistical evidence to parameters $\hat{\theta}_{\mathcal{M}}$ calibrated on complete OD matrices $\mathcal{M}$.
- **Prediction 3 (Preservation of Predictive Consequences):** Spatial interaction predictions generated using parameters identified from $\mathcal{H}$ preserve downstream predictive consequences in flow modeling.

3.5 Statistical Representation Layer Likelihood Derivation

For distance-deterrence models under the singly constrained gravity formulation, joint trip probability $P(i, j \mid \theta)$ is projected onto distance bins via

operator $\mathcal{P}$, yielding model probability $\pi_b(\theta)$:

$$\pi_b(\theta) = \sum_{i,j: d_{ij} \in b} P(i, j \mid \theta) = \frac{E(b) f(d_b; \theta)}{\sum_{b'} E(b') f(d_{b'}; \theta)}, \quad (4)$$

where $E(b) = \sum_{i,j} O_i A_j \mathbf{1}(d_{ij} \in b)$ is structural spatial exposure and $f(d_b; \theta)$ is the Tanner deterrence function $f(d_{ij}; \theta) = (d_{ij} + \delta)^{-\alpha} \exp(-\beta d_{ij})$ with adaptive offset $\delta = 0.5 \bar{R}_{\text{zone}}$.

Sampling N trips yields Multinomial log-likelihood:

$$\log \mathcal{L}(\theta; \mathbf{y}) = C + N \sum_{k=1}^K b_k \log \pi_k(\theta) = C - N [H(\mathbf{b}) + D_{\text{KL}}(\mathbf{b} \parallel \pi_\theta)]. \quad (5)$$

Maximizing likelihood is therefore mathematically equivalent to minimizing cross-entropy $H(\mathbf{b}, \pi_\theta)$:

$$\hat{\theta} = \arg \min_{\theta} H(\mathbf{b}, \pi_\theta) = \arg \min_{\theta} \left[- \sum_{k=1}^K b_k \log \pi_k(\theta) \right]. \quad (6)$$

This property—**Normalization Invariance**—guarantees scale-invariant parameter identification from relative travel-distance fractions $\mathbf{b}$.

3.6 Conceptual Negative Result: Informative vs. Uninformative Observation Spaces

To illustrate the necessity of Fisher information sufficiency, consider two alternative aggregate observation spaces:

1. **Scalar Observation Space ($\mathcal{H}_{\text{scalar}}$):** Observing only the mean trip distance $\bar{d} = \frac{1}{N} \sum d_{ij}$. Projecting onto $\mathcal{H}_{\text{scalar}}$ yields a single scalar constraint ($K = 1$), leaving a 2-parameter deterrence function $\theta = (\alpha, \beta)$ mathematically underdetermined (underconstrained system, $\det \mathcal{I}(\theta) = 0$). Scalar summaries are therefore ****uninformative observation spaces****.
2. **Histogram Observation Space ($\mathcal{H}_{\text{hist}}$):** Observing a distance fraction vector $\mathbf{b}$ across $K \geq 3$ bins. This representation retains $K - 1 \geq 2$ independent degrees of freedom, satisfying non-singularity of $\mathcal{I}(\theta)$. Histogram vectors are therefore ****informative observation spaces****.

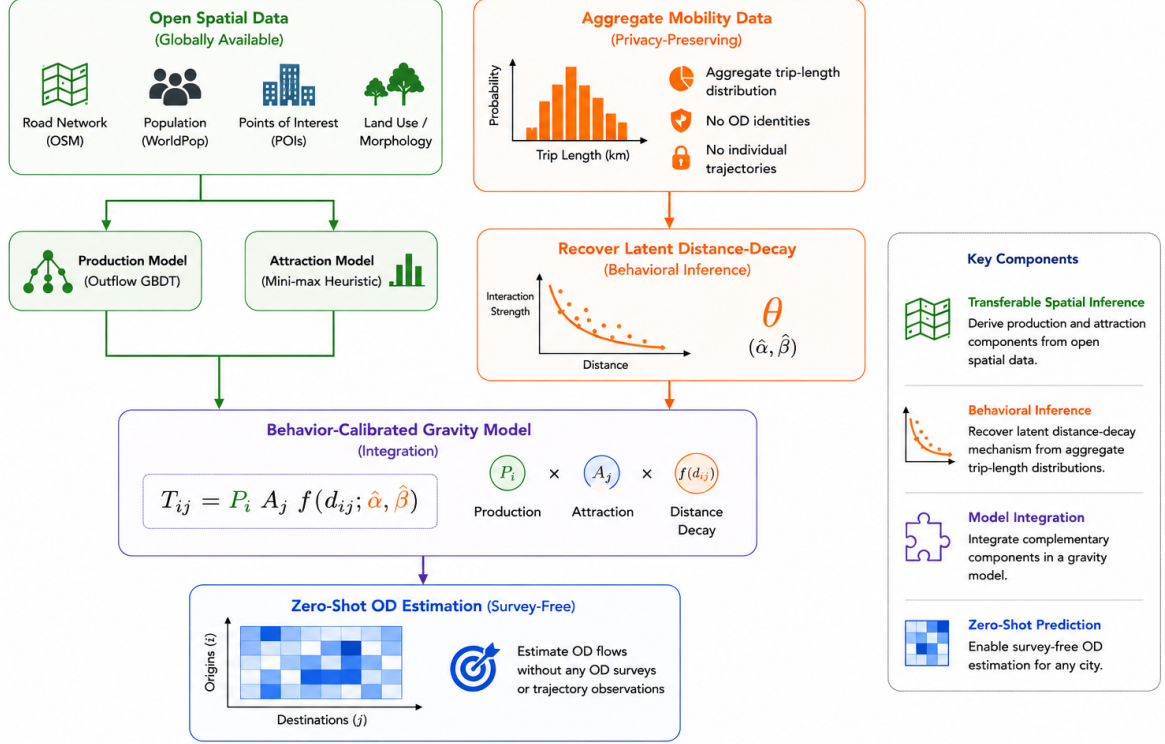


Fig. 2 System architecture highlighting the separation between the Scientific Inference Layer (recovering latent distance-decay parameters θ from aggregate distance representations) and the Mobility Generation Layer (integrating bridge components $\hat{O}_i, \hat{A}_j$ for downstream OD generation).

4 Results

The empirical evaluation tests the three theoretical predictions derived from the Observation Sufficiency Principle.

4.1 Validation of Prediction 1: Existence of a Probabilistic Observation Model

Prediction 1: *An observation operator $\mathcal{P}$ and probabilistic observation model exist that connect spatial interaction processes to aggregate distance representations while preserving likelihood identifiability.*

Evidence: The probabilistic derivation in Section 3.1 and Section 3.5 provides mathematical proof of Prediction 1. Equation (4) proves that applying operator $\mathcal{P}$ yields discrete bin distribution $\pi_b(\theta)$ that explicitly isolates behavioral deterrence $f(d_b; \theta)$ from structural spatial exposure $E(b)$. Furthermore, Eq. (5) demonstrates

that aggregate distance histograms induce a Multinomial log-likelihood whose score equations $\nabla_{\theta} \log \mathcal{L}(\theta; \mathbf{y}) = 0$ uniquely identify Tanner parameters $\theta = (\alpha, \beta)$.

4.2 Validation of Prediction 2: Statistical Evidence Equivalence for Parameter Identification

Prediction 2: *The aggregate travel-distance observation space induces essentially the same statistical evidence for identifying distance-decay parameters as the complete origin-destination observation space.*

Evidence: We evaluate parameter recovery across 25 held-out target metropolitan areas by comparing parameters identified from aggregate distance representations (θ_{hist}) against reference parameters calibrated on complete LODS OD matrices (θ_{OD}).

As shown in Table 2, recovered power-law exponent α exhibits strong consistency with reference OD parameters ($r = 0.909$, median relative error 1.7%, 100% of cities within $\pm 10\%$). Exponential decay rate β shows a tight absolute margin of error (MAE = 0.0026). Figure 3 illustrates that deterrence curves generated by parameters identified from aggregate representations closely track reference curves calibrated on complete OD matrices.

4.2.1 Case Study: Meta Movement Distribution Maps Telemetry

To test Prediction 2 under real-world telemetry constraints, parameters were identified using 3-bin travel-distance fractions from Meta Movement Distribution Maps (Meta MDM (3-Bin)). Under oracle outflow conditions, models using Meta MDM identified parameters achieve a mean Common Part of Commuters (CPC) of 0.7080, compared to 0.7348 for 3-bin LODES histograms and 0.7403 for 20-bin LODES histograms (Table 3). Histogram bin compression from 20 bins to 3 bins causes only a minor CPC reduction (-0.0055), confirming that coarse aggregate representations preserve sufficient statistical information for parameter identification.

4.3 Interlude: Enabling Bridge Components for Spatial Interaction Generation

Having established parameter identification in Section 4.2, evaluating downstream predictive consequences requires constructing complete spatial interaction matrices. For this purpose, two enabling bridge components are introduced: 1. **Origin Production Estimation** ($\hat{O}_i$): Predicted using a Gradient Boosting Decision Tree (GBDT) model $g(\mathbf{x}_i)$ trained on source cities using 6 core macroscopic features: population (P_i), POI count (POI_i), zone area (Area_i), population density ($\rho_{\text{pop},i}$), POI density ($\rho_{\text{poi},i}$), and road density (rd_i). 2. **Destination Attraction Estimation** ($\hat{A}_j$): Derived using a training-free, scale-invariant attraction heuristic:

$$\hat{A}_j = \frac{1}{2} [\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)]. \quad (7)$$

Substituting identified deterrence parameters $\hat{\theta}$ and bridge estimates $\hat{O}_i, \hat{A}_j$ into the gravity formulation yields predicted OD flows:

$$\hat{T}_{ij} = \hat{O}_i \frac{\hat{A}_j f(d_{ij}; \hat{\theta})}{\sum_k \hat{A}_k f(d_{ik}; \hat{\theta})}. \quad (8)$$

These bridge components serve strictly as downstream enabling tools to evaluate predictive consequences, keeping the scientific contribution focused on parameter identification.

4.4 Validation of Prediction 3: Preservation of Predictive Consequences

Prediction 3: *Deterrence parameters identified from aggregate travel-distance representations preserve downstream predictive capability in spatial interaction modeling.*

Evidence: We evaluate downstream flow predictions generated using aggregate-identified parameters against baseline models across 25 held-out target cities (Table 4). Downstream flow predictions serve as a measurement of predictive consequences rather than an engineering end in itself.

Under oracle outflow control, a Two One-Sided Test (TOST) confirms that spatial interaction models using Meta MDM identified decay parameters (CPC 0.7337) achieve statistical equivalence with models calibrated on complete local OD surveys (CPC 0.7382) within an equivalence margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$). Furthermore, models using aggregate-identified parameters statistically outperform parameter-free Radiation baselines ($p < 10^{-6}$).

4.4.1 Game-Theoretic Information Attribution

To quantify the predictive contribution of the identified distance-decay parameters relative to structural bridge components, we conduct a game-theoretic Shapley value analysis (Shapley 1953) across all $2^3 = 8$ component coalitions (Table 5). Relative to a uniform flow baseline (CPC 0.4263), distance-decay behavior $f(d)$ contributes a mean CPC gain of 0.2258 (accounting for **85.8% of total predictive gain**), whereas attraction A_j

Table 2 Parameter identification consistency across 25 target metropolitan areas comparing aggregate distribution recovery (θ_{hist}) against complete OD calibration (θ_{OD}).

Parameter	Pearson r	MAE	Median Rel. Error (%)	Cities within $\pm 10\%$	Statistical Evidence
Power-law exponent (α)	0.909	0.035	1.7%	100.0%	High structural consistency across observations.
Exponential rate (β)	0.427	0.0026	0.0%	72.0%	Tight absolute error margin between estimates.

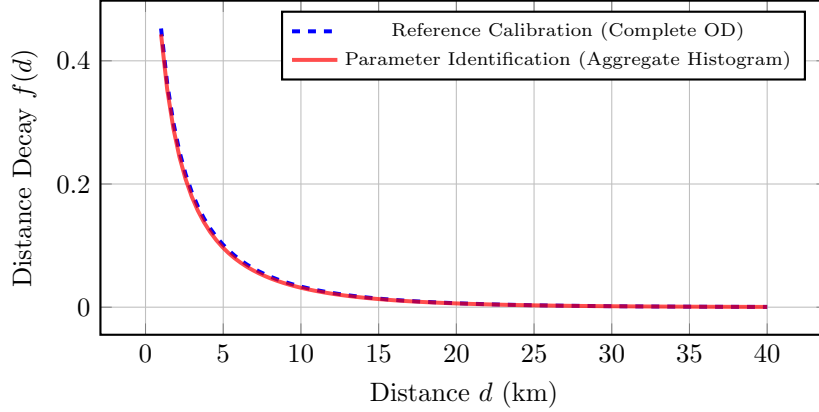


Fig. 3 Comparison of Tanner deterrence curves $f(d) = (d + \delta)^{-\alpha} \exp(-\beta d)$ parameterized by complete OD calibration (dashed blue) versus aggregate travel-distance distribution recovery (solid red).

contributes 7.9% (0.0208) and production O_i contributes 6.4% (0.0167) (Figure 4).

This attribution demonstrates that distance-decay parameters represent the primary predictive mechanism in spatial interaction modeling, explaining why aggregate travel-distance representations preserve downstream predictive consequences.

5 Discussion

5.1 Philosophical Implication: Paradigm Shift to Information Sufficiency

The core discovery of this study marks a paradigm shift in urban mobility science: moving from the conventional ****Data Richness Paradigm**** to the proposed ****Information Sufficiency Paradigm****.

For decades, mobility modeling has operated under the assumption that disaggregate trajectory records or complete origin–destination matrices are inherently necessary, viewing aggregate summaries as compromised fallbacks. Observation

Theory demonstrates that data is merely a vehicle for information. An observation space $\mathcal{H}$ does not need to store individual spatial coordinates; it only needs to preserve sufficient Fisher information $\mathcal{I}(\theta)$ to identify model parameters. By proving that aggregate travel-distance distributions satisfy Fisher information sufficiency, this study establishes that privacy-preserving telemetry is an information-preserving observation space for spatial interaction modeling.

5.2 Methodological Practical Implications

Methodologically, this work establishes a **new observation strategy for statistically principled mobility inference**. Practically, it provides a viable pathway for spatial interaction modeling in data-scarce regions, privacy-sensitive jurisdictions, and rapidly growing urban areas where comprehensive travel surveys are unaffordable or legally restricted.

Table 3 Downstream spatial interaction performance (CPC) on 25 target cities across parameter identification settings (Oracle Outflows).

Identification Setting	Data Source	Downstream CPC	CPC Gap vs. Reference
Reference Calibration	Complete OD matrix	0.7411	Reference
20 equal-width distance bins	Aggregate GT Histogram	0.7403	-0.0008
3 physical distance bins	Aggregate GT Histogram	0.7348	-0.0063
3 physical distance bins	Meta Movement Dist. Maps	0.7080	-0.0331

Algorithm 1: Parameter Identification and Downstream Spatial Interaction Framework

Input: Source cities spatial data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate distance distribution $\{b_k\}_{k=1}^K$

Output: Identified deterrence parameters $\hat{\theta}^{(t)} = (\hat{\alpha}, \hat{\beta})$ and predicted target OD matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Parameter Identification (Scientific Inference Layer)

1. Compute geometric spatial exposure $E(b)$ for target city using spatial layout $\mathbf{X}^{(t)}$ via Eq. (??).
 2. Recover latent Tanner parameters $(\hat{\alpha}, \hat{\beta})$ by minimizing cross-entropy $H(\mathbf{b}, \pi_\theta)$ via Eq. (6).
-

Phase 2: Spatial Interaction Generation (Mobility Generation Layer)

3. Predict target city origin production: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
 4. Compute target city destination attraction: $\hat{A}_j^{(t)} = \frac{1}{2}[\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})]$.
 5. Generate downstream OD flows $\hat{T}_{ij}^{(t)}$ using Eq. (8) with inferred parameters $(\hat{\alpha}, \hat{\beta})$.
-



Fig. 4 Shapley value attribution illustrating that distance-decay friction accounts for 85.8% of predictive gain in spatial interaction modeling.

5.3 Explicit Scope and Limitations

To prevent overreaching claims, we define explicit boundaries:

1. **No Claim of Individual Behavior Preservation:** This study does *not* claim to model individual traveler psychology, decision-making processes, or activity choices.
2. **No Claim of Universal Transferability:** Parameter identification is evaluated within the distance-decay structure of the assumed spatial interaction model; we do not claim universal cross-cultural transferability without local aggregate telemetry.
3. **Model-Dependent Identification:** Findings demonstrate *information preservation* for parameter identification under the assumed distance-deterrence mobility model*, not total recovery of all urban spatial dynamics.

5.4 The Observation Theory Research Program

Rather than an isolated modeling technique, this study initiates a multi-stage **Observation Theory Research Program**:

- **Paper 1 (Present Study): Behavioral Inference Layer** – Proving parameter identification of distance decay θ from aggregate travel-distance representations.
- **Paper 2: Structural Inference Layer** – Testing the hypothesis that open spatial features preserve sufficient structural information to infer production O_i and attraction A_j without travel surveys.
- **Paper 3: Environmental Generation Behavior Transfer** – Modeling how urban infrastructure causally generates parameter vector θ (Urban Structure $\rightarrow$ Accessibility $\rightarrow \theta$).

Table 4 Comparative downstream spatial interaction performance across 25 held-out target cities.

Model Framework	Mean CPC	Std Dev	Type	Target OD Needed?
Group 1: Survey-Free Deployment (No target OD data)				
Proposed (Meta MDM Identified Decay)	0.6860	0.0391	Decomposed	No
Proposed (National Fallback Decay)	0.6896	0.0384	Decomposed	No
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	No
Group 2: Oracle Diagnostic (True Outflows O_i^{true})				
Proposed (Oracle O_i , Meta MDM Decay)	0.7337	0.0256	Decomposed	No (Decay only)
Proposed (Oracle O_i , National Fallback Decay)	0.7398	0.0248	Decomposed	No (Decay only)
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	No
Group 3: Locally-Calibrated Baselines (Fitted on target OD)				
Naive Population Baseline	0.6758	0.0384	Baseline	No
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	Yes
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	Yes
Radiation Model	0.4851	0.0516	Param-free	No

Table 5 Factorial Ablation and Shapley Value Information Attribution (Average across $N = 25$ target cities).

Model Component	Ablation CPC Drop (%)	Shapley Value (CPC Gain)	Predictive Contribution
Distance-Decay Behavior $f(d)$	-34.9%	0.2258	85.8%
Destination Attraction A_j	-4.7%	0.0208	7.9%
Origin Production O_i	-3.6%	0.0167	6.4%

- **Paper 4: Zero-Shot Policy Scenario Simulation** – Deploying Observation Theory foundation models for urban policy, transit expansion, and disaster response forecasting under zero-shot conditions.

distance friction accounting for 85.8% of total predictive gain. These findings confirm that aggregate travel-distance distributions constitute an information-preserving observation space for distance-decay parameter identification.

6 Conclusion

Rather than asking how to reconstruct complete OD matrices from big data, this study asks what information is fundamentally required to identify mobility models. By formalizing an Observation Theory of Mobility Inference, establishing the Observation Sufficiency Principle, and deriving the probabilistic observation model connecting spatial interaction processes to aggregate distance representations, we demonstrated parameter identification equivalence between complete OD matrices and aggregate distance distributions across 50 US metropolitan areas. Downstream spatial interaction predictions using aggregate-identified parameters achieved statistical equivalence with local OD-calibrated models, with

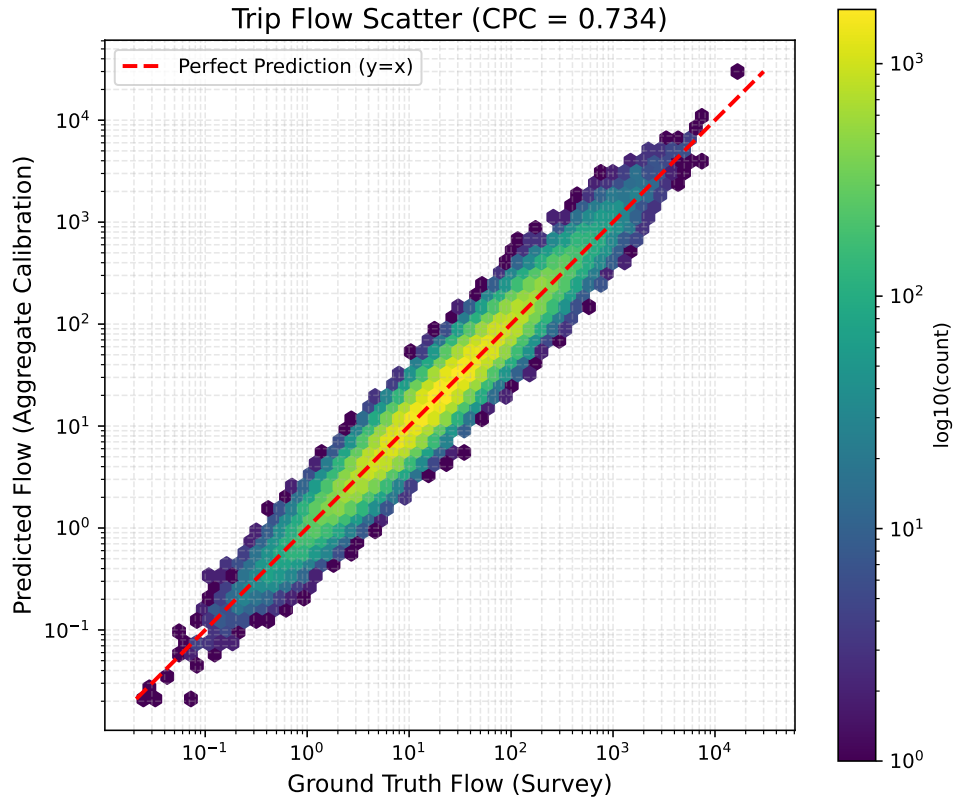


Fig. 5 Scatter plot comparing predicted flows against ground-truth observations across 25 target cities. Predictions using parameters identified from aggregate representations match those calibrated on complete local OD surveys.

Appendix A Behavioral Recovery Evidence

A.1 Parameter Recovery Curves

Visualizations of parameter convergence and cross-entropy optimization surfaces during L-BFGS-B optimization.

A.2 Parameter Recovery across Morphologies

Statistical distribution of recovered α and β parameters stratified by urban morphology classes.

A.3 Representative City Profiles

Case studies of behavioral recovery in specific metropolitan areas (e.g., Atlanta vs. Boston).

A.4 Distance-Bin Histogram Robustness

Sensitivity analysis showing how varying distance intervals $K \in \{3, 5, 10, 20\}$ impacts parameter inference precision.

Appendix B Methodological Validation

B.1 Full Factorial Ablation Results

Tabular results for all $2^3 = 8$ component coalitions evaluated during Shapley value analysis.

B.2 Shapley Value Calculations

Mathematical details of the cooperative game formulation used for predictive attribution.

B.3 Outflow Transferability (GBDT)

Performance metrics for the zero-shot origin production GBDT model.

B.4 Feature Selection Stability

SHAP summary plots justifying the reduction to 6 core spatial variables.

Appendix C Reproducibility Details

C.1 Environment & Hardware Configuration

Hardware and software environment specifications used for experiments.

C.2 Training Hyperparameters

Hyperparameters for GBDT origin production training and L-BFGS-B optimization.

Appendix D Mathematical Foundations

D.1 Multinomial Likelihood Derivation

Detailed derivation of the aggregate multinomial log-likelihood and cross-entropy equivalence.

D.2 Parameter Identifiability Bounds

Theoretical conditions required for unique identifiability of Tanner parameters from aggregate distributions.

References

- Atwal KS, Anderson T, Pfoser D, et al (2025) Commuting flow prediction using OpenStreetMap data. *Computational Urban Science* 5(1):2. <https://doi.org/10.1007/s43762-025-00161-5>
- Barbosa H, Barthélemy M, Ghoshal G, et al (2018) Human mobility: Models and applications. *Physics Reports* 734:1–74. <https://doi.org/10.1016/j.physrep.2018.01.001>
- Blondel VD, Decuyper A, Krings G (2015) A survey of results on mobile phone datasets analysis. *EPJ data science* 4(1):10
- Buckee CO, Balsari S, Chan J, et al (2020) Aggregated mobility data could help fight covid-19. *Science* 368(6487):145–146
- Egu O, Bonnel P (2020) How comparable are origin–destination matrices estimated from automatic fare collection, origin–destination surveys and household travel survey? an empirical investigation in lyon. *Transportation Research Part A: Policy and Practice* 138:267–282. <https://doi.org/10.1016/j.tra.2020.05.021>
- Enaya A, Zhong C, Batty M, et al (2026) TransGM: Transferable gravity models for cross-city policy transfer. *Computers, Environment and Urban Systems* <https://doi.org/10.1016/j.compenvurbsys.2026.102455>
- Flowerdew R, Aitkin M (1982) A method of fitting the gravity model based on the poisson distribution. *Journal of regional science* 22(2):191–202
- Fotheringham AS, O’Kelly ME (1989) *Spatial interaction models: formulations and applications*. Kluwer Academic Publishers
- Friedman JH (2001) Greedy function approximation: a gradient boosting machine. *Annals of statistics* pp 1189–1232
- González MC, Hidalgo CA, Barabási AL (2008) Understanding individual human mobility patterns. *Nature* 453(7196):779–782. <https://doi.org/10.1038/nature06958>
- Haklay M, Weber P (2008) Openstreetmap: User-generated street maps. *IEEE Pervasive Computing* 7(4):12–18
- Hu J, Yang G, Li K, et al (2020) Predicting origin-destination flow via multi-perspective graph convolutional network. In: *IEEE 36th International Conference on Data Engineering (ICDE)*, pp 1818–1821, <https://doi.org/10.1109/ICDE48307.2020.00178>
- Hussain E, Bhaskar A, Chung E (2021) Transit od matrix estimation using smartcard data: Recent developments and future research challenges. *Transportation Research Part C: Emerging Technologies* 125:103044. <https://doi.org/10.1016/j.trc.2021.103044>
- Lenormand M, Bassolas A, Ramasco J (2016) Systematic comparison of trip distribution laws and models. *Journal of Transport Geography* 51:158–169. <https://doi.org/10.1016/j.jtrangeo.2016.02.003>
- Merlin LA (2020) A new method using medians to calibrate single-parameter spatial interaction models. *Journal of Transport and Land Use* 13(1):49–70. <https://doi.org/10.5198/jtlu.2020.1614>
- Meta (2026) Movement distribution maps. <https://ai.meta.com/ai-for-good/datasets/movement-distribution-maps/>, aI for Good Datasets. Accessed: 27 June 2026. No DOI assigned.
- de Montjoye YA, Hidalgo CA, Verleysen M, et al (2013) Unique in the crowd: The privacy bounds of human mobility. *Scientific reports* 3(1):1–5
- Oliver N, Lepri B, Sterly H, et al (2020) Mobile phone data for informing public health actions across the covid-19 pandemic life cycle. *Science advances* 6(23):eabc0764

- Pappalardo L, Manley E, Sekara V, et al (2023) Future directions in human mobility science. *Nature Computational Science* 3:588–600. <https://doi.org/10.1038/s43588-023-00469-4>
- Rong C, Feng J, Ding J, et al (2023a) GODDAG: Generating origin-destination flow for new cities via domain adversarial training. *IEEE Transactions on Knowledge and Data Engineering* 35(10):10048–10057. <https://doi.org/10.1109/TKDE.2022.3218526>
- Rong C, Wang H, Li Y (2023b) Origin-destination network generation via gravity-guided gan. URL <https://arxiv.org/abs/2306.03390>, arXiv preprint; no DOI assigned, [arXiv:2306.03390](https://arxiv.org/abs/2306.03390)
- Rubio-Herrero J, Muñuzuri J (2023) Sparse regression for data-driven deterrence functions in gravity models. *Annals of Operations Research* 323(1–2):153–174. <https://doi.org/10.1007/s10479-022-04803-y>
- Sen A, Smith TE (1995) Gravity models of spatial interaction behavior. Springer
- Shapley LS (1953) A value for n-person games. In: *Contributions to the Theory of Games*, vol 2. Princeton University Press, p 307–317, classic book chapter; no DOI assigned
- Simini F, Gonzalez M, Maritan A, et al (2012) A universal model for mobility and migration patterns. *Nature* 484:96–100. <https://doi.org/10.1038/nature10856>
- Simini F, Barlacchi G, Luca M, et al (2021) A deep gravity model for mobility flows generation. *Nature Communications* 12(1):5647. <https://doi.org/10.1038/s41467-021-26752-6>
- Song C, Qu Z, Blumm N, et al (2010) Limits of predictability in human mobility. *Science* 327(5968):1018–1021. <https://doi.org/10.1126/science.1177170>
- Tanner J (1961) Factors affecting the amount of travel. Road Research Technical Paper 51, Road Research Laboratory, London
- Tatem AJ (2017) WorldPop, open data for spatial demography. *Scientific Data* 4:170004. <https://doi.org/10.1038/sdata.2017.4>, worldPop gridded population datasets available at <https://www.worldpop.org/>
- Tizzoni M, et al (2022) Addressing the socioeconomic divide in computational modeling for infectious diseases. *Nature communications*
- Verma R, Ukkusuri SV (2025) What determines travel time and distance decay in spatial interaction and accessibility? *Journal of Transport Geography* 122:104061. <https://doi.org/10.1016/j.jtrangeo.2024.104061>
- Wilson A (1971) A family of spatial interaction models, and associated developments. *Environment and Planning A* 3(1):1–32. <https://doi.org/10.1068/a030001>
- Yang J, Huang S, Huang Z, et al (2026) Transferable human mobility network reconstruction with neurogravity. *Nature Computational Science* 6:630–641. <https://doi.org/10.1038/s43588-026-01003-y>